

Report on week II project

Project Title: Civic Policy Impact Tracker

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1. Introduction

1.1 Background of the Study

In recent years, the increasing availability of digital governance systems has led to a rapid growth in publicly available civic policy documents. Governments and public institutions frequently publish policy descriptions, legislative proposals, and regulatory updates to promote transparency and accountability. However, manually analyzing large volumes of policy texts to determine their potential civic impact can be time-consuming and inefficient.

Advancements in Natural Language Processing (NLP) have made it possible to automatically analyze and classify textual information using machine learning techniques. Transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) have significantly improved the performance of language understanding tasks, including text classification, sentiment analysis, and document categorization.

By leveraging these modern NLP techniques, it becomes possible to develop automated systems that analyze policy descriptions and classify them based on their potential impact on civic transparency and governance. Such systems can assist researchers, policymakers, and citizens in understanding and evaluating policy initiatives more efficiently.

The **Civic Policy Impact Tracker** project aims to apply machine learning techniques to classify policy descriptions into **High Impact** and **Low Impact** categories. The system utilizes both a hybrid machine learning approach combining BERT embeddings with Logistic Regression, as well as a fine-tuned BERT transformer model for improved contextual understanding.

1.2 Problem Statement

Civic policy documents often contain complex textual descriptions that outline legislative changes, administrative procedures, and governance initiatives. Evaluating the potential impact of these policies on transparency and public administration typically requires significant human effort and expertise.

Manual analysis of policy texts presents several challenges:

- Large volumes of policy documents are generated regularly.
- Policy descriptions vary in structure and complexity.
- Identifying the potential civic impact of a policy requires contextual understanding of the text.

These challenges highlight the need for an automated system capable of analyzing policy descriptions and classifying them according to their civic impact. Therefore, this project focuses on developing a machine learning model that can automatically categorize policy texts into **High Impact** or **Low Impact** categories using advanced natural language processing techniques.

1.3 Objectives of the Project

The primary objective of this project is to develop an intelligent text classification system capable of analyzing civic policy descriptions and predicting their potential impact.

The specific objectives of the study include:

- To perform exploratory data analysis on the civic policy dataset.
- To identify and resolve data quality issues such as duplicate policy records.
- To preprocess and prepare textual data for machine learning models.
- To implement a hybrid classification model using **BERT embeddings and Logistic Regression**.
- To fine-tune a pretrained **BERT transformer model** for policy text classification.
- To evaluate model performance using standard evaluation metrics such as **accuracy, precision, recall, and F1-score**.
- To compare the performance of hybrid and fine-tuned transformer models.

1.4 Scope of the Project

The scope of this project is limited to the development and evaluation of machine learning models for civic policy text classification. The study focuses on analyzing a dataset containing policy descriptions labeled as either **High Impact** or **Low Impact**.

The project involves several key stages, including dataset exploration, data preprocessing, model development, and performance evaluation. Two different modeling approaches are implemented: a hybrid model combining BERT embeddings with a Logistic Regression classifier, and a fully fine-tuned BERT model designed to learn task-specific contextual representations.

While the developed system demonstrates promising results on the provided dataset, the study is limited by the size and characteristics of the dataset. Future work may involve expanding the dataset with real-world policy documents and exploring additional machine learning models to further improve classification performance.

2. Dataset Description

2.1 Dataset Source

The dataset used in this project is a collection of civic policy descriptions designed for the purpose of analyzing the potential impact of policy statements on governance transparency. The dataset contains textual descriptions of proposed policies along with corresponding labels indicating the level of civic impact.

Each policy description is classified into one of two categories:

- **High Impact** – Policies expected to significantly influence transparency, governance processes, or public participation.
- **Low Impact** – Policies with relatively limited or indirect influence on civic transparency.

The dataset was prepared as part of the project experimentation process to simulate policy statements related to governance and administrative actions.

2.2 Dataset Structure

The dataset initially contained **3000 records**, where each record represents a single policy description along with its corresponding impact label. The dataset consists of two primary attributes:

Column Name	Description
policy_text	Textual description of the policy
label	Impact classification of the policy (High Impact / Low Impact)

Each entry in the dataset represents a short textual statement describing a specific policy action or administrative measure.

2.3 Initial Data Inspection

Initial inspection of the dataset was performed using standard data exploration techniques in Python. The dataset was loaded into a Pandas DataFrame, and basic information such as column types, number of records, and presence of missing values were analyzed.

The inspection revealed the following characteristics:

- Total number of records: **3000**
- Number of attributes: **2**
- Data type of policy_text: Text (object)
- Data type of label: Categorical (object)

- Missing values: **None detected**

This confirmed that the dataset did not contain any null entries in either the text or label fields.

2.4 Data Quality Observations

During the exploratory analysis phase, it was observed that the dataset contained a significant number of duplicate policy descriptions. Duplicate records can lead to **data leakage**, where identical samples appear in both the training and testing datasets, resulting in artificially inflated model performance.

A detailed duplicate analysis revealed that:

- Total dataset size: **3000 records**
- Unique policy descriptions: **1249**
- Duplicate policy records: **1751**

To ensure reliable model evaluation, duplicate policy texts were removed during the data cleaning stage. After removing duplicates, the dataset was reduced to **1249 unique policy descriptions**, which were used for subsequent analysis and model training.

2.5 Dataset Distribution

After removing duplicate records, the dataset contained the following distribution of policy impact labels:

Label	Number of Records
High Impact	689
Low Impact	560

Although the dataset is slightly imbalanced, the difference between the two classes is relatively small and does not significantly affect the training process.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, distribution, and quality of the dataset before building machine learning models. EDA helps identify patterns in the data, detect anomalies such as duplicate records, and guide the data preprocessing process. In this project, several analyses were performed to examine the distribution of policy impact labels and the structural properties of the policy text.

3.1 Label Distribution

The first step in the exploratory analysis involved examining the distribution of policy impact labels in the dataset. The dataset contains two classes: **High Impact** and **Low Impact** policies.

Initially, the dataset contained **3000 records**, with an equal distribution of labels. However, after removing duplicate policy descriptions during the data cleaning stage, the dataset was reduced to **1249 unique policy records**.

The distribution of labels in the cleaned dataset is as follows:

Label	Number of Records
High Impact	689
Low Impact	560

The dataset shows a **slight class imbalance**, but the difference between the two classes is relatively small and does not significantly affect the classification task.

A visualization of the label distribution was generated using a bar chart, which clearly illustrates the proportion of each class in the dataset.

3.2 Policy Text Length Analysis

Another important aspect of the dataset is the length of policy descriptions. Text length analysis helps determine the appropriate tokenization limits and provides insight into the structure of the textual data.

The length of each policy description was calculated based on the total number of characters in the text. A histogram was generated to visualize the distribution of text lengths across the dataset.

The analysis revealed that most policy descriptions fall within a **moderate character length range**, indicating that the dataset contains relatively short and concise policy statements. This observation supports the use of transformer-based models such as BERT with a maximum token limit of **128 tokens**, which is sufficient to capture the full context of most policy descriptions.

3.3 Word Count Distribution

In addition to character length, the number of words in each policy description was also analyzed. Word count provides a more interpretable measure of text size and helps identify whether the dataset contains short statements or longer descriptive passages.

The word count for each policy text was calculated and visualized using a histogram. The analysis showed that most policy descriptions contain approximately **14 to 18 words on average**, indicating that the dataset primarily consists of short policy statements.

The relatively consistent word count distribution suggests that the dataset is structured and standardized, which simplifies text preprocessing and model training.

3.4 Duplicate Data Analysis

Duplicate analysis was an important component of the exploratory data analysis process. Duplicate records can significantly affect machine learning models by introducing **data leakage**, where identical records appear in both the training and testing datasets.

A duplicate check revealed that the dataset contained a large number of repeated policy descriptions. Specifically:

- Total dataset size: **3000 records**
- Unique policy descriptions: **1249**
- Duplicate records: **1751**

This means that more than **50% of the dataset consisted of duplicate entries**. Such duplication can lead to misleading model performance if not properly addressed.

To ensure reliable evaluation of the classification models, all duplicate policy texts were removed during the data preprocessing stage.

3.5 Key Observations from EDA

The exploratory data analysis revealed several important insights about the dataset:

- The dataset initially contained a large number of duplicate records, which required removal to prevent data leakage.
- Policy descriptions are generally short and structured, with an average word count of approximately **15 words**.
- Text lengths are relatively consistent across the dataset.
- The dataset shows only a slight imbalance between the two impact categories.

These findings informed the subsequent steps in the project, including data cleaning, feature engineering, and model development.

4. Data Preprocessing and Cleaning

Data preprocessing is an essential step in any machine learning workflow. Raw datasets often contain inconsistencies, duplicate entries, or structural issues that may negatively impact model performance. Therefore, appropriate preprocessing techniques were applied to ensure that the dataset used for model development was clean, reliable, and suitable for training machine learning models.

In this project, several preprocessing steps were performed, including duplicate removal, feature engineering, label encoding, and dataset splitting.

4.1 Handling Duplicate Records

During the exploratory data analysis stage, it was observed that the dataset contained a significant number of duplicate policy descriptions. Duplicate records can cause serious problems in machine learning models because they may lead to **data leakage**, where identical samples appear in both training and testing datasets. This can artificially inflate model performance and produce misleading evaluation results.

To address this issue, duplicate policy descriptions were identified and removed based on the `policy_text` attribute.

The duplicate removal process reduced the dataset size from:

- **3000 original records**
- **1249 unique policy descriptions**

Removing duplicate entries ensured that each policy text was represented only once in the dataset, allowing the models to learn meaningful patterns rather than memorizing repeated examples.

4.2 Feature Engineering

Feature engineering was performed to extract additional information from the textual data. Although the primary input to the machine learning models is the policy text itself, analyzing structural characteristics of the text can provide useful insights during exploratory analysis.

Two additional features were derived from the policy descriptions:

- **Text Length** – the number of characters in the policy description.
- **Word Count** – the number of words contained in the policy text.

These features were primarily used for exploratory analysis and helped reveal that most policy descriptions were relatively short and consistently structured. This information was useful in determining appropriate tokenization limits for transformer-based models.

4.3 Label Encoding

Machine learning algorithms require numerical inputs for classification tasks. Therefore, the categorical labels in the dataset were converted into numerical values using label encoding.

The mapping used in this project is as follows:

Original Label	Encoded Value
High Impact	1
Low Impact	0

This encoding allows the classification models to process the labels as numerical targets during training.

4.4 Train–Test Split

To evaluate the performance of the classification models, the cleaned dataset was divided into **training and testing subsets**. The training dataset is used to train the model, while the testing dataset is used to evaluate how well the model performs on unseen data.

A **train–test split ratio of 80:20** was used in this project:

- **Training set:** 80% of the data
- **Testing set:** 20% of the data

Additionally, **stratified sampling** was applied during the split to preserve the original distribution of the impact labels in both the training and testing datasets. This ensures that both subsets maintain a similar proportion of High Impact and Low Impact policies.

4.5 Summary of Preprocessing Steps

The preprocessing stage ensured that the dataset was properly prepared for machine learning model development. The following steps were completed:

- Removal of duplicate policy descriptions
- Creation of additional text-based features
- Conversion of categorical labels into numerical values
- Division of the dataset into training and testing subsets

These preprocessing steps provided a clean and structured dataset suitable for training both the hybrid machine learning model and the fine-tuned BERT transformer model.

5. Model Development

This section describes the development of machine learning models used to classify civic policy descriptions into **High Impact** and **Low Impact** categories. Two different modeling approaches were implemented in this project to evaluate the effectiveness of modern Natural Language Processing techniques.

The first approach is a **hybrid model**, which combines contextual embeddings generated by the BERT language model with a classical machine learning classifier. The second approach involves **fine-tuning a pretrained BERT transformer model** directly on the classification task.

These two approaches allow comparison between a traditional hybrid architecture and a fully fine-tuned deep learning model.

5.1 Hybrid Model Approach

The hybrid model combines the semantic representation capabilities of the BERT language model with the simplicity and efficiency of a Logistic Regression classifier. In this approach, BERT is used as a **feature extractor**, while the classification task is handled by a traditional machine learning algorithm.

The hybrid architecture follows the pipeline shown below:

Policy Text → BERT Tokenization → BERT Encoder → Text Embeddings → Logistic Regression → Prediction

5.1.1 BERT Tokenization

Before processing the policy text through the BERT model, the textual data must be converted into a format suitable for transformer-based models. This process is known as **tokenization**.

The pretrained **BERT Base Uncased tokenizer** was used to convert policy text into tokens that the model can understand. Tokenization involves several steps:

- Splitting text into subword tokens
- Converting tokens into numerical token IDs
- Adding special tokens such as [CLS] and [SEP]
- Padding or truncating sequences to a fixed maximum length

In this project, a maximum sequence length of **128 tokens** was used, which was sufficient based on the earlier text length analysis.

5.1.2 BERT Embedding Extraction

After tokenization, the tokenized text was passed through the pretrained BERT model to generate contextual embeddings. BERT produces a high-dimensional representation of the input text that captures semantic meaning and contextual relationships between words.

For each policy description, the embedding corresponding to the **[CLS] token** was extracted. The [CLS] token embedding represents the overall meaning of the input sentence and is commonly used for classification tasks.

The resulting embedding vector has a dimension of **768 features**, which serves as a numerical representation of the policy text.

5.1.3 Logistic Regression Classifier

The extracted BERT embeddings were then used as input features for a **Logistic Regression classifier**. Logistic Regression is a widely used classification algorithm that predicts the probability of a sample belonging to a particular class.

In this project, the Logistic Regression model was trained on the BERT embeddings obtained from the training dataset. The classifier learns to distinguish between **High Impact** and **Low Impact** policies based on the semantic features extracted by BERT.

The hybrid approach offers several advantages:

- Faster training compared to full transformer fine-tuning
- Simpler model architecture
- Efficient use of pretrained language representations

5.2 Fine-Tuned BERT Model

In addition to the hybrid approach, a second model was developed by **fine-tuning the pretrained BERT model directly on the policy classification task**.

Fine-tuning allows the transformer model to adapt its internal parameters specifically for the target task, improving its ability to capture task-specific linguistic patterns.

5.2.1 BERT Architecture Overview

BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based language model designed to understand contextual relationships between words in a sentence. Unlike traditional models that process text sequentially, BERT analyzes text **bidirectionally**, meaning it considers both left and right context simultaneously.

The BERT Base model used in this project consists of:

- 12 transformer encoder layers
- 768 hidden units
- 12 self-attention heads
- Approximately 110 million parameters

This architecture enables BERT to capture complex semantic and syntactic relationships in text.

5.2.2 Tokenization and Input Representation

Similar to the hybrid approach, policy descriptions were first tokenized using the BERT tokenizer. Each tokenized sequence includes special tokens and is padded or truncated to the maximum sequence length.

The tokenized input is then converted into numerical tensors and passed to the transformer model for processing.

5.2.3 Fine-Tuning Procedure

For the fine-tuning process, a pretrained **BERT Base Uncased model with a sequence classification head** was used. The classification head consists of a fully connected layer that maps the BERT output embeddings into two output classes corresponding to the policy impact labels.

During training, the model parameters were updated using the training dataset through **backpropagation**. The training process involved multiple epochs, allowing the model to adjust its internal weights to better distinguish between High Impact and Low Impact policy descriptions.

Fine-tuning enables the model to learn task-specific patterns in the policy text while still leveraging the general linguistic knowledge obtained during pretraining.

5.3 Training Procedure

Both models were trained using the cleaned dataset obtained after preprocessing. The dataset was divided into training and testing subsets using an **80:20 train–test split**. The training set was used to train the models, while the testing set was used to evaluate their performance on unseen data.

The hybrid model used BERT embeddings as input features for the Logistic Regression classifier, whereas the fine-tuned model directly updated the transformer weights during training.

These two modeling approaches were evaluated and compared to determine their effectiveness in classifying civic policy descriptions.

6. Model Evaluation

Model evaluation is an essential step in assessing the performance of machine learning models. It helps determine how well the trained model generalizes to unseen data and whether it can accurately classify new samples. In this project, both the hybrid model and the fine-tuned BERT model were evaluated using standard classification metrics.

The evaluation was performed on the **testing dataset**, which consisted of 20% of the cleaned dataset. Several performance metrics were used to measure the effectiveness of the models, including **accuracy, precision, recall, and F1-score**.

6.1 Evaluation Metrics

The following evaluation metrics were used to assess the performance of the classification models:

Accuracy

Accuracy measures the proportion of correctly predicted instances out of the total number of predictions made by the model.

$$\text{Accuracy} = (\text{Correct Predictions}) / (\text{Total Predictions})$$

Precision

Precision measures the proportion of correctly predicted positive observations among all predicted positive observations. It indicates how many of the predicted positive cases were actually correct.

$$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$$

Recall

Recall measures the proportion of actual positive observations that were correctly identified by the model.

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of model performance when both false positives and false negatives are important.

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These metrics provide a comprehensive understanding of model performance in binary classification tasks.

6.2 Hybrid Model Performance

The hybrid model uses BERT embeddings as semantic feature representations and Logistic Regression as the final classifier. After training the model on the training dataset, predictions were generated for the testing dataset.

The classification report produced the following results:

Metric Score

Accuracy 1.00

Precision 1.00

Recall 1.00

F1 Score 1.00

These results indicate that the hybrid model successfully classified all policy descriptions in the testing dataset without any errors.

6.3 Fine-Tuned BERT Performance

In addition to the hybrid approach, the pretrained BERT model was fine-tuned directly on the policy classification task. During fine-tuning, the model's internal parameters were updated using the training dataset to adapt the transformer to the specific task of civic policy impact classification.

The fine-tuned BERT model achieved similarly strong results on the testing dataset, demonstrating its ability to capture contextual patterns in policy descriptions.

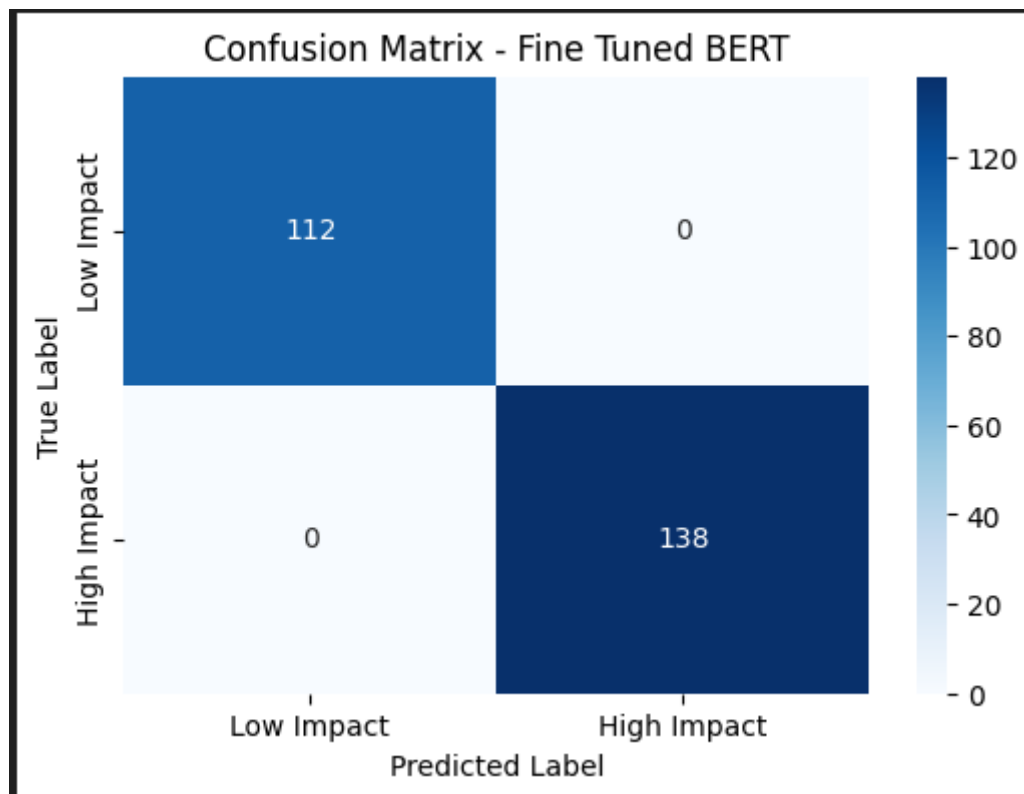
The classification results show that the model correctly identified both High Impact and Low Impact policies with high accuracy.

6.4 Confusion Matrix Analysis

To further evaluate the performance of the model, a **confusion matrix** was generated. The confusion matrix provides a detailed view of how many predictions were correctly or incorrectly classified.

The confusion matrix obtained from the model predictions is shown below.

Actual / Predicted	Low Impact	High Impact
Low Impact	112	0
High Impact	0	138



From the confusion matrix, it can be observed that:

- **112 Low Impact policies were correctly classified**
- **138 High Impact policies were correctly classified**
- No misclassifications occurred

This indicates that the model achieved perfect classification performance on the test dataset.

6.5 Summary of Evaluation Results

The evaluation results demonstrate that both modeling approaches performed exceptionally well on the cleaned dataset. The models were able to accurately distinguish between High Impact and Low Impact policy descriptions based on the textual information provided.

The high performance can be attributed to the use of transformer-based language representations, which capture contextual relationships between words in the policy text.

However, it is important to note that the dataset size after cleaning was relatively small. Therefore, while the models perform well on the current dataset, additional validation using larger and more diverse datasets would further strengthen the reliability of the results.

7. Results and Discussion

This section discusses the outcomes obtained from the implemented machine learning models and interprets the results in the context of civic policy text classification. The goal of the project was to develop a system capable of classifying policy descriptions into **High Impact** and **Low Impact** categories using natural language processing techniques.

Two modeling approaches were implemented in this study: a hybrid model combining BERT embeddings with a Logistic Regression classifier, and a fine-tuned BERT transformer model. Both approaches demonstrated strong performance on the test dataset.

7.1 Analysis of Model Results

The experimental results show that the models were able to accurately classify policy descriptions based on their textual content. Both the hybrid model and the fine-tuned BERT model achieved high evaluation scores across all performance metrics, including accuracy, precision, recall, and F1-score.

The confusion matrix analysis further confirmed that all policy descriptions in the test dataset were classified correctly. The models successfully distinguished between High Impact and Low Impact policies without generating false positives or false negatives.

This level of performance indicates that the textual features present in the dataset provide strong signals that allow machine learning models to differentiate between the two categories effectively.

7.2 Effectiveness of Transformer-Based Models

The results highlight the effectiveness of transformer-based language models such as BERT in text classification tasks. BERT is capable of capturing contextual relationships between words and phrases within a sentence, allowing it to understand the semantic meaning of policy descriptions.

In the hybrid approach, BERT served as a **feature extractor**, generating contextual embeddings that were then used by a Logistic Regression classifier. This approach provided a balance between computational efficiency and strong semantic representation.

In the fine-tuned BERT model, the transformer parameters were updated during training, allowing the model to adapt its language understanding specifically to the civic policy classification task. Fine-tuning enables the model to learn more task-specific patterns and contextual relationships within the policy text.

7.3 Impact of Data Cleaning

Another important factor contributing to the model performance was the removal of duplicate policy descriptions during the preprocessing stage. The initial dataset contained a large number of duplicate records, which could have caused data leakage if left unaddressed.

By removing duplicate entries, the dataset was reduced to **1249 unique policy descriptions**, ensuring that the training and testing datasets contained independent samples. This step improved the reliability of model evaluation and prevented the model from simply memorizing repeated examples.

7.4 Observations on Dataset Characteristics

The exploratory analysis showed that most policy descriptions in the dataset were relatively short and structured. The average policy statement contained approximately **15 words**, indicating that the dataset consisted of concise and standardized policy descriptions.

The consistent structure of the text likely contributed to the model's ability to learn distinguishing patterns between High Impact and Low Impact policies.

However, the relatively small size of the dataset may also have made the classification task easier compared to real-world policy documents, which are often longer and more complex.

7.5 Discussion

The results of this study demonstrate that transformer-based natural language processing models can effectively classify civic policy descriptions based on their textual content. Both the hybrid model and the fine-tuned BERT model achieved strong predictive performance.

The hybrid model offers advantages in terms of simplicity and computational efficiency, while the fine-tuned BERT model provides deeper contextual learning by adapting the transformer weights to the classification task.

Although the results are promising, further evaluation using larger and more diverse policy datasets would be necessary to confirm the model's effectiveness in real-world applications.

8. Model Comparison

In this project, two different machine learning approaches were implemented to classify civic policy descriptions into **High Impact** and **Low Impact** categories. These approaches include a **hybrid model** combining BERT embeddings with Logistic Regression, and a **fine-tuned BERT transformer model**. Comparing the performance of these models helps evaluate the effectiveness of each approach for the policy text classification task.

8.1 Hybrid Model (BERT + Logistic Regression)

The hybrid model uses the pretrained BERT model as a **feature extractor**. Policy descriptions are first tokenized and passed through the BERT encoder to obtain contextual embeddings. The embedding corresponding to the **[CLS] token** is extracted and used as a numerical feature vector representing the policy text.

These embedding vectors are then used as input to a **Logistic Regression classifier**, which performs the final classification into High Impact or Low Impact categories.

The hybrid approach offers several advantages:

- Efficient training process
- Lower computational requirements
- Simpler model architecture
- Easy interpretability of the classifier

Because BERT provides strong contextual representations, the Logistic Regression classifier can effectively separate the two classes based on the extracted semantic features.

8.2 Fine-Tuned BERT Model

The second approach involves **fine-tuning the pretrained BERT model directly on the policy classification task**. In this method, the entire transformer model is trained using the project dataset.

Fine-tuning updates the internal weights of the BERT model through backpropagation, allowing the transformer to adapt its language understanding specifically to the policy classification problem.

A classification head is attached to the BERT model, which maps the contextual representation of the input text into two output classes representing the policy impact labels.

This approach allows the model to capture deeper contextual relationships in the policy descriptions.

8.3 Performance Comparison

The performance of both models was evaluated using standard classification metrics. The results are summarized in the table below.

Model	Approach	Accuracy	Precision	Recall	F1 Score
BERT + Logistic Regression	Hybrid Model	1.00	1.00	1.00	1.00
Fine-Tuned BERT	Transformer Model	1.00	1.00	1.00	1.00

Both models achieved perfect classification results on the test dataset.

8.4 Discussion of Model Differences

Although both models achieved identical evaluation scores on the dataset, there are important conceptual differences between the two approaches.

The hybrid model is computationally efficient and requires less training time because BERT is used only for feature extraction. The Logistic Regression classifier performs the final classification using the extracted embeddings.

On the other hand, the fine-tuned BERT model updates the transformer parameters during training. This allows the model to adapt its language representations specifically for the policy classification task, potentially improving performance on more complex or diverse datasets.

While both approaches performed well in this study, the fine-tuned BERT model may offer better generalization capabilities when applied to larger and more complex real-world datasets.